



Research paper

An integrated Multi-Regional Input-Output (MRIO) Analysis of miscanthus biomass production in France: Socio-economic and climate change consequences



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ABSTRACT

Several European policies have been designed over the last decades to address the challenge of climate change and several measures have been put in place to accelerate the development and deployment of cost-effective low carbon technologies. The domestic nature of the resource and its great potential availability in Europe make biomass conversion technologies relevant mitigation options to be considered. In this context, the project “Logistics for Energy Crops Biomass (LogistEC)” aims to develop new or improve technologies of biomass logistics chain. In this project, the sustainability of different types of biomass is analysed in terms of environmental, economic and social impacts, based on the supply chain of two existing plants. The objective of this paper is to present the main results obtained in the socio-economic analysis of the French case and its climate change consequences. The Input-Output Analysis (IOA) has been seen as the most appropriate method to estimate these impacts using a Multiregional Input-Output Table from the World Input-Output Database project. Socio-economic effects have been estimated in terms of additional economic activity, added value and job creation. By extending the IOA with environmental accounts, greenhouse gas (GHG) emissions have also been estimated. Additionally, the most stimulated sectors have been identified. Results highlight the importance of biomass at a national level.

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1. Introduction

The EU members states adopted in 2009 the 2020 climate and energy package [1–3] which is a part of the Europe 2020 strategy for smart, sustainable and inclusive growth [4]. Under this strategy, the European Union aims to reduce its GHG emissions by at least 20%, increase the share of renewable energy to at least 20% of consumption, and achieve energy savings of 20% or more [3,4]. All EU countries must also achieve a 10% share of renewable energy in their transport sector.

Recently, European countries have agreed on a new 2030 Framework for climate and energy, including EU- targets and policy objectives for the period between 2020 and 2030. These targets aim to help the EU achieve a more competitive, secure and sustainable energy system and to meet its long-term 2050 GHG reductions

target. [5].

Different strategies are required to attain the energy system transformation required to meet these ambitious goals. In order to accomplish a sustainable energy system, while meeting the GHG reductions targets, the European Strategic Energy Technology plan (SET-plan) [6], communicated in 2007, promotes the development of competitive and affordable low-carbon technologies.

Bioenergy is expected to play an important role towards this energy transition. Biomass is a domestic resource, available in many European regions that can be stored to generate the energy on demand [7]. However, there are still many barriers to overcome for biomass to be deployed to a large scale. The European Innovation Partnership Agricultural Productivity and Sustainability (EIP-AGRI) identified the main barriers for the development of bioenergy, and proposed tools and measures to overcome them (EIP-AGRI, 2015). The technical difficulties, the cost of biomass logistics, the economic challenges due to price fluctuations and competition from other uses as well as the farmers' scepticism are some of the most mentioned [8].

Additionally, and due to the domestic nature of the resource,

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bioenergy can also promote local economy and employment. These positive socioeconomic impacts should be taken into account when designing a successful bioenergy policy. To trigger its relevance in the future energy mixes, this resource should be cost-efficient, environmental-friendly and socially sustainable, besides economic and environmental efficient. The Logistics for Energy Crops Biomass (LogistEC) project [9], funded by the 7th Framework Programme (FP7), aims to develop new or improve technologies of biomass logistics chain by considering the environmental, economic and social aspects of sustainability. The project focuses on improving all biomass value chain components. It analyses different types of biomass for two existing plants. To include the three pillars of sustainability, the analyses are conducted under a common framework that enables, firstly, an economic optimization based on feedstock models, predictions of the most probable location of future fields and distances, and then, the environmental and socio-economic assessment considering the different life cycle stages from “cradle to gate”. The environmental analyses are carried out following the Life Cycle Assessment method, while the socio-economic analysis is based on Multi-regional Input-Output (MRIO) Analysis.

In this paper, we present the socio-economic impacts from the production of energy from miscanthus at Bourgogne Pellets plant, located in the Burgundy region of Eastern France. Although it is out of the project scope, we have also extended the MRIO Analysis with environmental accounts to estimate the GHG emissions associated to the system.

This paper is organized as follows: we describe the method in section 2. Section 3 presents the case study and main hypothesis. Section 4 discusses the results of the study. Finally, we draw the main conclusions in section 5.

2. Method

The methodology used to estimate the socio-economic effects of the production of bioenergy of miscanthus is the Input-Output (IO) Analysis, extended with sectorial employment vectors as well as sectorial GHG emissions vectors and considering the trade between the different economies of the world.

IO Analysis has been commonly used to estimate macro-economic impacts of industries within the national or regional economy. This analysis was developed by Wassile Leontief for which he was awarded with the Nobel Prize in Economics in 1973. IO analysis describes, through symmetrical tables, the interdependencies between activity sectors within an economy [10,11]. The IO tables are built based on statistical data for a given economy that could be at regional or national level and describe the flows of goods and services in monetary terms between different activity sectors and demands or between producers and consumers. The IO tables can be divided into two main components, the inter-industry flows or transaction matrix, which describes the flows from sector i to sector j and the final demand. Intermediate goods and services are further processed by other sectors [12].

An IO table represents the production cost components in columns, accounting for the resources consumed from other sectors to obtain a certain production in each sector, while in rows the distribution of one sector's production among the other sectors is described. Table 1 shows an example of an input-output table.

Therefore, the total output from each sector is defined by:

$$x_i = z_{i1} + z_{i2} + \dots + z_{in} + y_i \quad (1)$$

This equation will be set for all sectors included in the IO table. We can use the matrix notation to describe the different elements of the equations system as:

$$x = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix}; Z = \begin{bmatrix} z_{11} & \dots & z_{1n} \\ \vdots & \ddots & \vdots \\ z_{n1} & \dots & z_{nn} \end{bmatrix}; y = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} \quad (2)$$

where x is a vector that expresses the total output, Z is the IO matrix and y is the final demand vector.

The IO matrix can be expressed with the technical coefficients, which represent the ratio of input to output, that is the amount required by one sector from other sector to produce one monetary unit of output. The technical coefficients are denoted as:

$$a_{ij} = z_{ij}/x_j \quad (3)$$

The technical coefficients can be expressed as a matrix, the A matrix. We substitute z_{ij} in Equation (1) for the technical coefficients and the total output is defined by the following matrix equation:

$$x = Ax + y \quad (4)$$

Reorganizing Equation (4), we get the following expression:

$$x = (I - A)^{-1}y \quad (5)$$

where $(I - A)^{-1}$ is the Leontief inverse matrix, or the multiplier matrix, that expresses the total production of each sector required to satisfy the final demand. That is the direct and indirect requirements per unit of final demand.

Through the IO analysis, we can estimate the total output in monetary terms that will be produced by the different sectors in the economy in order to satisfy the intermediate and final demand of goods and services. Equation (5) can be also decomposed as follows:

$$x = (I + A + AA + AAA + \dots + A^n)y \quad (6)$$

where Iy are the direct impacts and $Ay + AAy + \dots + A^ny$ are the indirect impacts.

The IO analysis began as a method to analyze national or regional economies. However, production processes have become less domestic, and national economies are part of a global economy. Supply chains are increasingly fragmented across borders and this fundamentally modifies the nature of international trade with important consequences for the location of production as well as other related impacts. MRIO modeling provides the opportunity to analyze the consequences of this fragmentation in a comprehensive way by including different regions and their trade relationships.

The MRIO tables follow a similar structure than the IO tables, but there are interregional and intraregional transactions, being

Table 1
Example of input-output table.

		Processing sectors (intermediate demand)				Final demand	Total output
		1	2	...	n		
Processing sectors	1	z_{11}	z_{12}	...	z_{1n}	Y	x
	2	z_{21}	z_{22}	...	z_{2n}		
	...	z_{31}	z_{32}	...	z_{3n}		
	n	z_{n1}	z_{n2}	...	z_{nn}		
Payment sectors	Value added	v_1	v_2	...	v_n		
	Import	m_1	m_2	...	m_n		
Total outlays		x					

Source: [11].

possible to identify the purchases that one sector in region 1 makes from other sector in region 2.

Currently, there are several MRIO databases available to conduct global analysis. OECD produced a database that comprises IO tables for 48 countries, covering 37 sectors. This database covers years from 2002 to 2006 [13]. EORA, developed by Lenzen and his team, is a MRIO database that works in native classification of each country. It includes data for 187 countries and 25 sectors [14]. Other well-known database is the EXIOBASE, a global MRIO database, which covers 43 countries and 5 “rest of the world” regions [15]. The reference year is 2007.

The choice of one database or other could influence the final results. Moran and Wood analyzed the reliability of different global MRIO databases, comparing the results of four of them, showing that the degree of disagreement varies by country and by model [16].

We have considered three criteria to choose the database: (i) the year of reference, (ii) the number of regions and countries and (iii) the number of activity sectors described.

Based on these criteria, we have chosen the World Input-Output Database (WIOD) to carry out this study. At the time when this study was designed, WIOD appeared as the best option among the others, covering the three main issues relevant for the study. The WIOD provides time-series -from 1995 to 2011- of world IO tables for 40 countries worldwide and a model for the rest-of-the-world, describing 35 activity sectors [17]. It was developed in a FP7 project. This database contains detailed information on national production activities for 27 EU countries, 13 major non-EU countries and 1 region referred to rest of the world. All data used in the WIOD have been obtained from official national statistics, mainly from National Accounts published by the National Statistical Institutes. Moreover, the WIOD also provides satellite accounts, necessary to extend the MRIO.

But IO analysis has not been only used to calculate impacts on the economy. In the early seventies, already Leontief, among others, extended the analysis to cover environmental aspects within the IO analysis [18,19]. Nowadays, the Environmentally Extended Input-Output (EEIO) Analysis is widely used and it is constantly under improvement through the hybridization with other methods such as Life Cycle Assessment [20–23].

To include any extension, environmental or social, we need to have an additional matrix or vector that provides the amount of pollutants, i.e., emitted by each activity sector per monetary unit of output. Including this vector into Equation (5), we get the final expression that we have used in this study:

$$e_i = l_i(I - A)^{-1}y \quad (7)$$

where l_i is the vector describing the direct impacts coefficients (per unit of output) and e_i is the total impact, direct and indirect, associated to the total output that satisfies the final demand.

The WIOD provides data on labour and capital inputs as well as on pollution indicators at the industry level that can be simultaneously used to enlarge the scope of possible applications including the estimation of job creation and some environmental impacts. In this study, besides the economic impacts we have included job creation and GHG emissions.

Although IO Analysis is not a novel methodology, its use in the energy sector has increased in the last years, particularly in biofuels production.

Kulisic et al. [24] performed an IO Analysis, where they estimated the impacts from biodiesel produced in Croatia, including this activity as a new industry in the national table. Neuwahl et al. estimated the employment consequences of the EU biofuels policy using an IO framework [25], considering prices effects and potential

household budget reduction among other variables. Mukhopadhyaya and Thomassin included two new industries and four new commodities in the Canadian Make and Use Table to estimate the macroeconomic effects of bioethanol in Canada [26]. [27] analysed and compared employment generation and economic effects on GDP through an IO Analysis of three types of bioethanol and palm oil biodiesel produced in Thailand. The Joint Research Centre published in 2014 a report that evaluates the macroeconomic impacts of four bio-based applications –biofuels, biochemical, bioelectricity, and biogas –using a general equilibrium model in the European Union [28]. Recently, Song et al. have developed a dynamic simulation model using an IO approach to evaluate energy, environmental and economic impacts of renewable energy policies scenarios introducing six energy conversion technologies, as well as on bioenergy scenarios [29,30].

3. Case study

3.1. *Miscanthus* production in Bourgogne

Bourgogne Pellets (BP) is a farmers' cooperative of about 350 farmers located in the municipality of Aiserey, 20 km south-east of Dijon (Eastern France). The plant has been specialized in the production of granules from sugar beet pulp during the last 40 years. However, it was in 2008 when the European Union decided to lower the sugar production quotas in France and BP had no other option than finding new resources to keep the activity.

BP began the production of pellets from wood (coniferous) and perennial grasses in 2008. Currently, it grows 320 ha of miscanthus and 60 ha of switchgrass. Due to the relatively high miscanthus production costs and the pellets low price in the market, BP has moved to other market and it currently sells its biomass for animal bedding.

Miscanthus (*Miscanthus × giganteus*) is a large, perennial, rhizomatous grass with high rates of growth [31]. These properties make miscanthus the only interesting grass from the bioenergy perspective. In the analyzed region, the stems can reach more than 4 m in only one season.

The sustainability of miscanthus is determined, among others, by the economic profit, and therefore by the spatial location of this crop. Using a Boosted Regression Trees (BRT) modeling approach, realistic prospective miscanthus field locations were identified in a previous step based on real spatial distribution data [32].

As mentioned previously, transport and logistics are some of the barriers to the development of bioenergy due to its high costs. Knowing the fields where miscanthus could grow, and with the aim of improving the supply chain logistics, an economic optimization model was used to determine decisions regarding harvesting, transport and storage [33]. Transport costs as well as storage were calculated based on the shortest paths using the road network of France. Loading and unloading costs were also included in the transport cost. The different percentage of chips, bales and pellets are based on the current mix used at BP (see Table 3).

3.2. Hypothesis and main data

This paper presents the analysis of the baseline scenario. We have considered an annual miscanthus production level of 6000 t(dm), based on the prospective arable areas and yields. This scenario considers around 400 ha available to grow miscanthus, and an annual average yield of 15 t(dm) per ha. Miscanthus cycle includes 14 harvestings. Table 2 presents the main agricultural activities assumed during the miscanthus production cycle.

Part of the harvested biomass is chipped in the field and transported directly to the BP plant. The remaining biomass is

Table 2
Miscanthus cycle main cultivation activities.

	1st year	2nd year	3rd to 16th year
Soil preparation	Ploughing, Cultivator	—	—
Planting (rhizomes/ha)	19000	—	—
Fertilizing (kg N/ha y)	59,4	59,4	59,4
Plant protection (l/h)	6	2	—
Harvesting	—	—	15
(average yield tdm/ha y)	—	—	—
Baling	—	—	Chips round bales square bales

Table 3
Baseline scenario definition.

Biomass	Baseline scenario (t/year)	Lower heating value (GJ/t)
Pellets	4944	16
Short bales	720	15,6
Chips	336	14,7

harvested in long and short bales and transported later to temporal storages and to BP, considering a maximum distance of 70 km. The following table (Table 3) shows the distribution between chips and bales followed to define the scenario.

The MRIO analysis will calculate the socio-economic and environmental impacts associated to a final demand of goods and services, denoted as y in the previous equations. Thus, it is necessary to quantify all goods required along the analysed system and to identify the sectors that produce and supply them in the economy to build the final demand vectors y . In this case study, the final good demanded by BP is miscanthus, which would be produced and supplied by the sector *Agriculture, Forest and Fishing*. However, this sector also produces many other goods, such as cereal crops, oil-seeds, rice, pork, etc, since the Input-Output tables aggregate different products into one unique sector. Demanding the amount of miscanthus crop required by BP directly from the *Agriculture, Forest and Fishing* sector, in the MRIO analysis would produce only “average” socio-economic and environmental impacts. This problem of sector aggregation is one of the limitations of IO analysis also in its extensions, and it has been studied by several authors [34–35]. With the objective of minimizing sector aggregation problems, we have done a step back, defining all goods and services required to get our final product, pellets from miscanthus at BP plant. This assumption considers intermediate requirements as final goods.

The final demand vectors describe the costs in monetary units required to cultivate, harvest, chip and bale miscanthus as well as its transport and storage and the final treatment to get pellets (Fig. 1). Based on these phases, we have divided the analysis in several stages:

- Machinery and other investments, which includes the machinery costs, insurances and maintenance costs. The type of machinery required in each agricultural operation has been described along the project based on the results of the demonstration work packages. In order to calculate these costs, we have used an on-line tool¹ developed by the Spanish Ministry of Agriculture, Food and Environment, based on the recommended methodology described in Ref. [36].

- Agricultural operations that comprise the application of fertilizers and other agrochemicals, as well as their transport to the farmers, the rhizomes used in the first year, the consumption of fuel for the agricultural operations and other costs such as loan interests. Most data have been directly provided by the farmers' cooperative Dijon Céréales.
- Bales and chips transport and storage, considering the distances and costs per ton resulting from the economic optimization model.
- Pelletizing, considering the equipment and consumption of goods, such as energy consumed in the operation.

Through the MRIO analysis we can also estimate other effects in the economy. The salaries earned by the workers will be partially spent in other goods and services, generating a new final demand and therefore, new impacts. This impact is defined as induced impact. In order to account for it, we have estimated the salaries and wages as well as the social contribution based on the working hours required in each phase and the data published by Eurostat from the French Agriculture sector in 2011 [37]. Social contribution has been included as part of the agricultural operation costs, and only the net salary has been used to estimate the induced effects. We have assumed that the workers have a tendency to save of 5% of the net salary and they demand goods and services following the average expenditure of final demand described in the French IO table in 2011.

4. Results and discussion

We present first the results of the economic performance of pellets production from miscanthus. Then, we describe the socio-economic and environmental impacts associated to the system. Finally, we identify the sectors and countries that contribute the most to each impact category.

4.1. Economic performance of miscanthus biomass

Based on the activities required along the whole process and the associated costs, we have calculated the economic performance of miscanthus biomass.

Table 4 shows the aggregated costs of the main phases distinguished previously. In order to consider the time effect, we have calculated the net present value of all costs and revenues along the whole miscanthus cycle considering a discount rate of 4%.

More than 50% of the total costs occur during the pelletizing stage, since it is a high energy demanding process. Agricultural operations take the second place, 26%, while transport and storage contributes with 12% to total costs, minimized by the use of the shortest paths. Considering all years of the crop cycle, from the rhizomes implantation until the moment when miscanthus is removed, and under the hypothesis described previously, producing 1 ton of biomass in form of chips, bales and pellets costs around 100 €.

With this information, together with the description of all the activities involved along the crop cycle, we have defined the final demand vectors, and have used them to calculate the socio-economic and environmental effects associated to the system. In total, more than ten activity sectors will provide the goods and services required directly by the final demand.

4.2. Miscanthus socio-economic impacts

We have estimated the socio-economic impacts associated to the miscanthus biomass production at BP plant in 400 ha, during the whole crop cycle. Table 5 presents the total impacts from each

¹ <http://www.magrama.gob.es/es/ministerio/servicios/informacion/plataforma-de-conocimiento-para-el-medio-rural-y-pesquero/observatorio-de-tecnologias-probadas/maquinaria-agricola/hojas-calculo-maqui.aspx>.

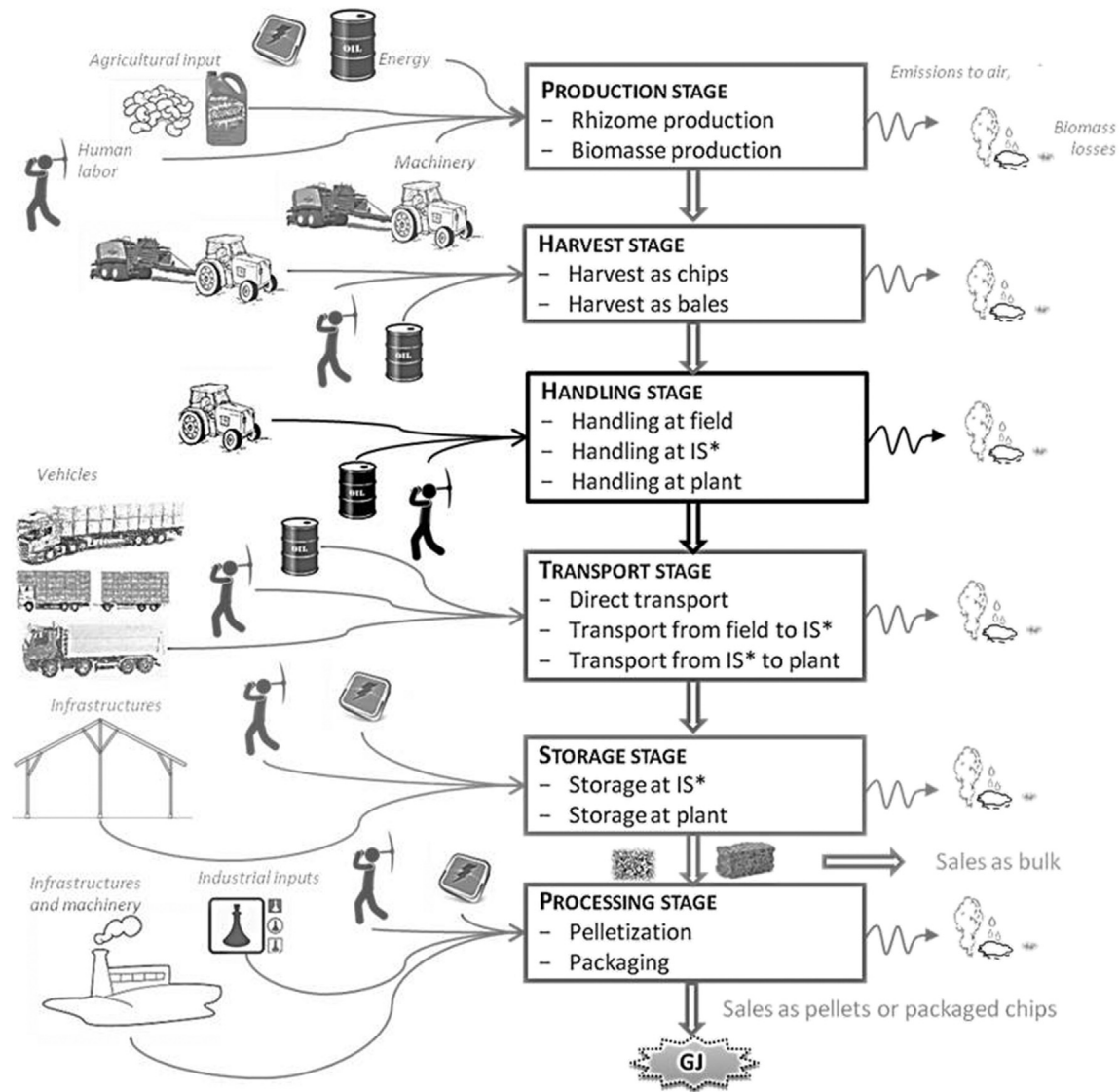


Fig. 1. Activities included in the analysis of miscanthus biomass production.

Table 4
Miscanthus biomass economic performance.

Stages	Costs	
	€/t	€/ha
Machinery & other investments	8.71	130.58
Labour operations	26.92	403.83
Transport & storage	12.30	184.48
Pelletizing	54.80	821.94
Total	102.72	1540.83

Table 5
Miscanthus socio-economic impacts.

Stages	Production of goods & services	Value added	Employees
	Million €	Million €	People
Machinery & other investments	1.13	0.44	8
Labour operations	3.60	1.26	24
Transport & storage	1.32	0.61	10
Pelletizing	8.23	2.60	49
Total	14.29	4.90	91

stage.

In order to satisfy the final demand of all goods and services required along all life cycle stages, from miscanthus sowing until the gate at BP, the global economy will produce more than 14 million € in goods and services, from which almost 5 million € contribute as value added to the GDP. This economic activity increase will generate 91 jobs measured as full-time equivalent. Pelletizing has the highest contribution to all impacts, accounting for more than 50% of the total production of goods and services and job creation, followed by the labour operations during miscanthus

production. When adding the socio-economic impacts caused by the induced effect, total production of goods and services rises to 14.5 million €.

Using the MRIO analysis we can divide the total effects in three categories: (i) direct impacts, caused by the direct demand of goods and services related to our system, (ii) indirect impacts, as the upstream economic activities needed to supply the direct goods and services, and (iii) induced impacts, caused by the partial spending of the net salary earned from the direct employees. Fig. 2

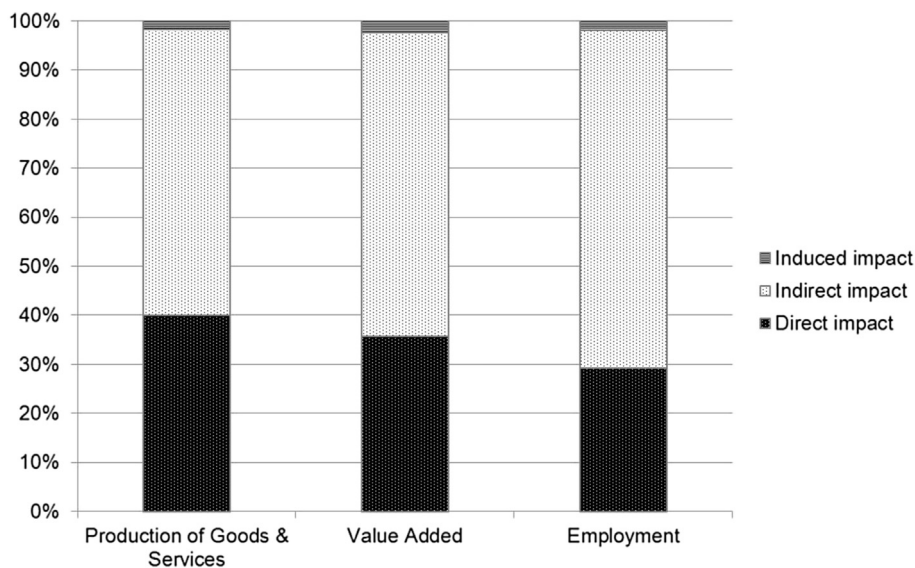


Fig. 2. Distribution of direct, indirect and induced impacts per category.

shows the relative importance of these three impacts in each socio-economic effect.

The induced impacts have the lowest contribution in the three categories, while the indirect impacts contribute more than 50% to all categories. In the case of employment, indirect impacts generate 70% of the jobs.

Using these results, it is possible to estimate the multiplier effect, which measures how an increase in one economic activity can cause an increase in the other activities in the whole economy. The multiplier effect is the ratio between the total economic impact estimated, denoted as X in the previous equations, and the amount of money directly spent in the biomass cycle in terms of goods and services, denoted as y in the equations. The multiplier effect obtained in this analysis is 2.44. This means that when the final consumers require 1 € of miscanthus biomass at the gate of BP

plant, there will be a total production of goods and services in the whole economy equivalent to 2.44 €. These results demonstrate that the studied bioenergy activity has a very important potential to boost the economy as a whole. These results can probably be extrapolated to other bioenergy projects in France and probably also in other countries with similar economic structure.

Bioenergy intends not only to reduce GHG emissions but also to contribute to the domestic economy and employment. The use of MRIO analysis offers the possibility to identify which effects occur within the domestic boundaries. Fig. 3 shows, in percentage, the socio-economic benefits that will occur in France, in other European countries and in other regions around the world.

Most of the economic benefits take place in France, which will gain more than 75% of production of goods and services and 70% of value added, as it could be expected since bioenergy activities make

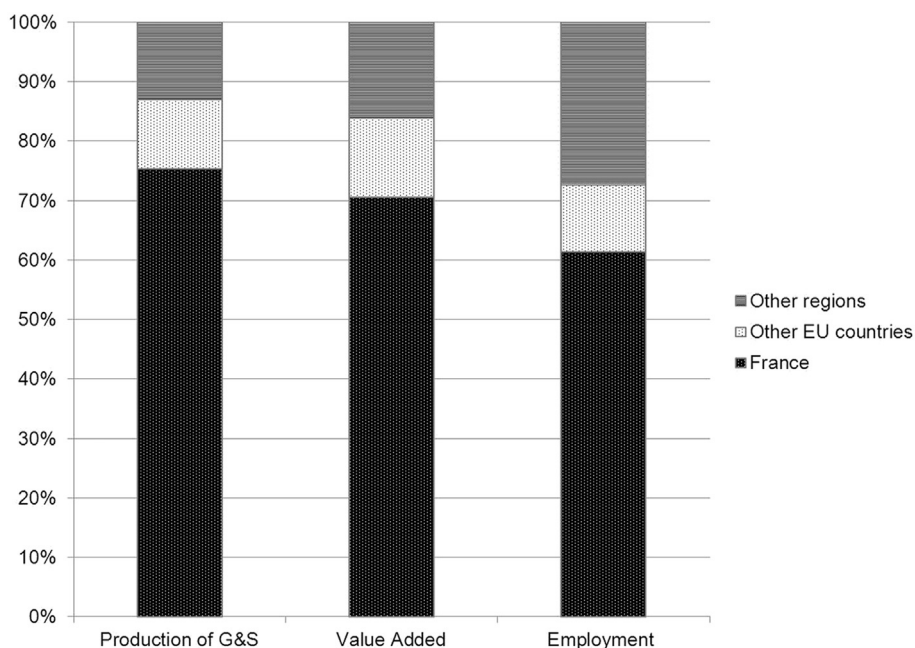


Fig. 3. Distribution of socio-economic impacts among the regions.

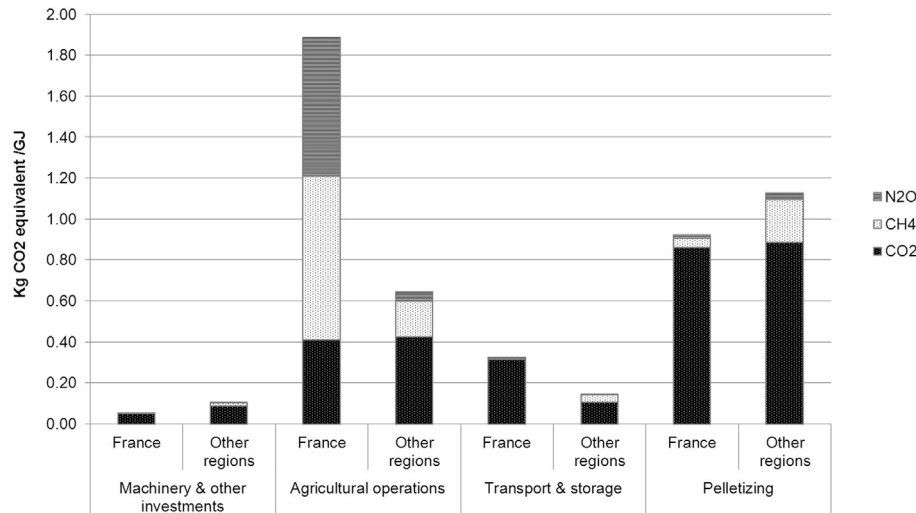


Fig. 4. Distribution of GHG per stage and regions.

use of domestic resources. However, other European countries will profit indirectly from the French miscanthus production as well. Additionally, around 12% of the total economic impacts will be produced in other countries outside Europe. The case of employment merits special attention, since it shows that there is not a linear relationship between value added and job creation. Around 15% of the value added is located in other regions outside the European boundaries, and these countries will also generate almost 30% of the total employment. This could be due to a higher ratio of employment per unit of economic activity in the different sectors stimulated by this activity in other countries of the world.

4.3. Miscanthus climate change impacts

Based on the environmental satellite accounts included in the WIOD, we have calculated climate change impact, measured as CO₂ equivalent, from three air pollutants: CO₂, CH₄ and N₂O. Following the same structure as in the previous subsection, Table 6 shows the results for the three pollutants for the whole system.

Results show both direct and indirect GHG emissions, being direct emissions those related to the direct production of goods and services defined by the final demand vectors, while indirect emissions are all emissions produced in the sectors supplying these sectors with goods and services. Based on this, direct emissions include those related to activities such as fuel consumption during agricultural operations, agrochemicals production, machinery manufacturing, transport or electricity consumption at BP plant. N₂O emissions from the direct application of fertilizers in the field have been included using the methodology provided by the IPCC Guidelines for National Greenhouse Gases Inventory [38]. Land use change emissions have been excluded as well as.

Table 6
Miscanthus environmental impacts.

Stage	Tons CO ₂ equivalent			GHG kg CO ₂ eq/GJ
	CO ₂	CH ₄	N ₂ O	
Machinery & other investments	125.64	16.62	3.67	0.16
Labour operations	755.02	871.35	1047.76	2.98
Transport & storage	381.30	38.23	4.38	0.47
Pelletizing	1573.30	227.85	41.59	2.05
Total	2835.25	1154.04	699.24	5.66

Considering 14 harvestings and the production of bales, chips and pellets, GHG emissions exceed 5000 tons of CO₂ equivalent, from which 56% are due to CO₂, 23% to CH₄ and 21% to N₂O. CO₂ equivalents are calculated considering the relative global warming potentials of the three gases in a 100-year period published by the IPCC [39]. The table also presents the emissions in kg of CO₂ equivalent per GJ of final biomass, which amount to a total of 5.66 kg of CO₂ equivalent per GJ. Direct emissions only represent 32% of the total, being most of the impact embedded in the upstream supply chain. Direct application of fertilizers in the field generates 7.8% of the total GHG, in form of N₂O emissions. Parajuli et al., estimate through a life cycle assessment the emission of 5 kg of CO₂ equivalent per MJ of miscanthus biomass used in a boiler [40]. Our results are higher than Parajuli's, as expected, since MRIO analysis does not define system boundaries.

The distribution of these emissions along the regions follows a similar trend as in the socio-economic impacts. Most of the GHG emissions are consequences of production activities located in France, 64% of the total. The indirect production from European countries contributes with 8% to the total, while other regions around the world will emit 28% of the total. Fig. 4 shows the distribution of each pollutant between France and the rest of regions in each stage of the system.

In some stages of the analysed system, global warming impacts occurring outside France are more important than the domestic ones. This is the case of the CO₂ emissions of the machinery production and other investments production and the CH₄ emissions in the pelletizing phase.

4.4. Sectorial analysis

One of the strengths of MRIO analysis is that using this method, we can identify not only the regions in which the impacts occur but also the activity sectors that have the highest contributions. Considering that we are using 40 countries, 1 region covering "Rest of the World" and 35 sectors in each region, we have highlighted those activity sectors that contribute more than 1% to each impact category. Fig. 5 indicates these results.

France is the main responsible for both the socio-economic benefits as well as the environmental impacts. From the socio-economic perspective, four sectors will benefit from miscanthus biomass production: (i) Agriculture, Forestry and Fishing, (ii) Electrical and Optical Equipment, (iii) Renting of machinery and other

COUNTRY	ACTIVITY SECTOR	Production G&S %	Value Added %	Employment %	GHG %
CHINA	Agriculture, Forestry and Fishing			2,35	
	Electrical and Optical Equipment	1,06		1,92	
	Electricity, Gas and Water Supply				2,96
	Other Social and Personal Services			1,60	
FRANCE	Agriculture, Forestry and Fishing	8,11	6,09	9,96	39,86
	Coke, Refined Petroleum and Nuclear Fuel	3,05			2,83
	Chemicals and Chemical Products	1,99			1,03
	Basic Metals and Fabricated Metal	3,23	2,83	1,84	1,59
	Machinery, Nec	2,89	2,51	1,90	
	Electrical and Optical Equipment	16,11	13,67	8,50	
	Electricity, Gas and Water Supply	9,21	3,39	2,93	13,29
	Sale, Maintenance of Vehicles	1,48	2,71	2,45	
	Wholesale Trade	2,76	2,87	2,25	
	Retail Trade	1,55	3,78	3,43	
	Inland Transport	5,83	9,06	7,89	6,18
	Other Auxiliary Transport Activities	1,31	1,35	1,13	
	Financial Intermediation	2,96	2,32	2,08	
	Real Estate Activities	1,09			
	Renting of Machinery & other businesses	8,17	11,58	9,92	
	Public Admin; Social Security		1,06		
RUSSIA	Mining and Quarrying				2,97
	Electricity, Gas and Water Supply				2,60
	Wholesale Trade		1,06		
	Inland Transport				1,27
RoW	Agriculture, Forestry and Fishing				1,05
	Mining and Quarrying	1,59			1,61
	Chemicals and Chemical Products				1,15
	Electricity, Gas and Water Supply				1,84
TOTAL COVERED WITH THESE SECTORS		72,39	64,28	60,16	80,22

Fig. 5. Activity sectors with contribution higher than 1% to each category.

businesses, and (iv) *Inland transport*. China will stimulate the employment in *Agriculture, Forestry and Fishing* sector mainly. The analysed system will trigger Russian *Wholesale trade* sector in terms of value added.

Analysing the sectorial results from the environmental results, three sectors located in France cover more than half of the total GHG emissions. These are: (i) *Agriculture, Forestry and Fishing*, (ii) *Electricity, Gas and Water Supply*, and (iii) *Inland transport*. Chinese *Electricity, Gas and Water Supply* sector is also responsible for almost 3% of the GHG emissions. Nowadays, China is one of the main suppliers of goods and services in the world. Manufacturing industries involve often high energy demanding processes. The environmental regulations in non-European countries are usually more moderate, or even non-existent. Thus, it is not surprising that this sector emerges as an important contributor to Miscanthus GHG emissions. The same case arises in Russia with other sectors such as *Mining and Quarrying* or *Electricity, Gas and Water Supply*.

This effect of delocalization of both environmental and socio-economic impacts is highly dependent on the use of imported goods by the analysed economic activity (the production of pellets, chips and bales of Miscanthus in this case study) but also by the degree of imports of the economy of the country where the investments are taking place (the French economy in this case study). Also the off-shoring of the economic activities of French companies could affect the distribution of the impacts. In order to generalize these results it is important to take these aspects into account. The same economic activity located in a country whose economy relies much more on imported goods than the French economy will have a higher share of both impacts and benefits located outside its boundaries.

5. Conclusions and recommendations

The work presented in this paper has estimated the

environmental and socio-economic effects of the production of pellets, chips and bales of *Miscanthus* in France. The results obtained have shown how this activity has an important potential to boost the national economy of France in terms of total economic production and value added but also can influence other economies. The multiplier effect calculated is of 2.44. This means that by each euro spent in the cycle 1.44 euros are additionally produced in the global economy, being the total production 2.44. Analyzing the results from a geographical perspective, more than 70% of the socio-economic benefits accrue within France. Outside France, due to import leakage, not only by the French economy but by Europe in general, Chinese sectors are the most indirectly benefitted from the BP production. Especially relevant is the employment creation in China as a consequence of the activity investigated. Most of the GHG emissions are located in France, but a significant part 36% are located outside the French boundaries as a consequence of the offshoring of the production processes in many sectors of the economy.

In France, the sectors most stimulated by the investigated process are the ones related to agricultural production, electrical equipment, renting of machinery and transport. The countries that would benefit the most from the production of *Miscanthus* in France would be China and Russia.

Most of the GHG emissions are produced in France although the energy production related sectors of China and Russia also contribute in a significant amount.

Although the socio-economic impacts estimated in this research reflect a very specific case study, these results could easily be generalized to other bioenergy projects in France with similar agriculture cycle and therefore similar cost structure. Crops that would comply with these limitations could be for example *switchgrass* (*Panicum virgatum*) or elephant grass (*Pennisetum purpureum*). Results would be less generalizable if the analyzed crops are very different from the one considered in this study (rhizomatous, perennial and herbaceous crop) or if the production processes are located in a country with a very different economic structure compared to the French economy.

What is, however, fully general is the method used to estimate the impacts. Once the environmental and socioeconomic vectors of the tool are defined and linked to the MRIO matrix, the methodology can very easily be used to estimate the environmental and socioeconomic sustainability performance of any other bioenergy chain, provided we know the demand vector generated by the different activities involved.

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References

- [1] Directive 2009/29/EC of the European Parliament and of the Council of 23 April 2009 amending Directive 2003/87/EC so as to improve and extend the greenhouse gas emission allowance trading scheme of the Community.
- [2] Directive 2012/27/EU of the European Parliament and of the Council of 25 October 2012 on energy efficiency, amending Directives 2009/125/EC and 2010/30/EU and repealing Directives 2004/8/EC and 2006/32/EC.
- [3] Directive 2009/28/EC of the European Parliament and of the Council of 23 April 2009 on the promotion of the use of energy from renewable sources and amending and subsequently repealing Directives 2001/77/EC and 2003/30/EC.
- [4] Communication from the Commission. Europe 2020. A strategy for smart, sustainable and inclusive growth. COM(2010) 2020 final. EUROPEAN COMMISSION Brussels, 3.3.2010.
- [5] Communication from the Commission to European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions A policy framework for climate and energy in the period from 2020 to 2030 (SWD (2014) 15 final) {SWD(2014) 16 final}COM(2014) 15 final.
- [6] Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions of 22 November 2007 entitled: "A European strategic energy technology plan (SET Plan) - towards a low carbon future" [COM(2007) 723 final].
- [7] Fereshteh Mafakheri, Fuzhan Nasiri, Modeling of biomass-to-energy supply chain operations: applications, challenges and research directions, Energy Policy 67 (C) (2014) 116–126.
- [8] Athanasios A. Rentizelas, Athanasios J. Tolis, Ilias P. Tatsiopoulos, Logistics issues of biomass: the storage problem and the multi-biomass supply chain, Renew. Sustain. Energy Rev. 13 (4) (2009) 887–894, 1364-0321.
- [9] LogistEC project. <http://www.logistecproject.eu>.
- [10] W. Leontief, The Structure of American Economy: 1919-1939, Oxford University Press, New York, 1941.
- [11] R.E. Miller, P.D. Blair, Input-output Analysis: Foundations and Extensions, second ed., Cambridge University Press, Cambridge, 2009.
- [12] K. Feng, K. Hubacek, A multi-region input-output analysis of global virtual water flows, in: M. Ruth (Ed.), Handbook of Research Methods and Applications in Environmental Studies, Edward Elgar Publishing Ltd, 2015.
- [13] N. Yamano, N. Ahmad, in: The OECD Input-output Database: 2006 Edition" OECD Science, Technology and Industry Working Papers, OECD Publishing, 2006, <http://dx.doi.org/10.1787/308077407044>.
- [14] M. Lenzen, D. Moran, K. Kanemoto, A. Geschke, E.O.R.A. Building, A global Multi-Region Input-Output database at high country and sector resolution, Econ. Syst. Res. 25 (1) (2013).
- [15] A. Tukker, A. de Koning, R. Wood, T. Hawkins, S. Lutter, J. Acosta, J.M. Rueda Cantuche, M. Bouwmeester, J. Oosterhaven, T. Drosowski, J. Kuenen, EXIO-POL – development and illustrative analyses of a detailed global MR EE SUT/IOT, Econ. Syst. Res. 25 (1) (2013) 50–70.
- [16] D. Moran, R. Wood, Convergence between the EORA, WIOD, EXIOBASE, and OPENEU's Consumption-based Carbon Accounts, Economic Systems Research, 2014.
- [17] M.P. Timmer, E. Dietzenbacher, B. Los, R. Stehrer, G.J. de Vries, An illustrated user guide to the world input–output database: the case of global automotive production, Rev. Int. Econ. (2015), <http://dx.doi.org/10.1111/roie.12178>.
- [18] Walter Isard, Some notes on the linkage of the Ecological and Economic systems, J. Reg. Sci. Assoc. Int. 22 (1) (1969) 85–96.
- [19] Wassily Leontief, Environmental Repercussions and the Economic Structure: an Input-output Approach the Review of Economics and Statistics, 52(3), MIT Press, 1970, pp. 262–271, <http://dx.doi.org/10.2307/1926294>.
- [20] C. Hendrickson, a Horvath, S. Joshi, L. Lave, Economic input-output models for environmental life-cycle assessment, Environ. Sci. Technol. 32 (7) (1998) 184a–191a.
- [21] Suh SI, M. Lenzen, G.J. Treloar, H. Hondo, A. Horvath, G. Huppes, O. Joliet, U. Klann, W. Krewitt, Y. Moriguchi, J. Munksgaard, G. Norris, System boundary selection in life-cycle inventories using hybrid approaches, Environ. Sci. Technol. 1–38 (3) (2004) 657–664.
- [22] Wiedmann TO1, S. Suh, K. Feng, M. Lenzen, A. Acquaye, K. Scott, J.R. Barrett, Application of hybrid life cycle approaches to emerging energy technologies—the case of wind power in the UK, Environ. Sci. Technol. 1–45 (13) (2011) 5900–5907, <http://dx.doi.org/10.1021/es2007287>. Epub 2011 Jun 8.
- [23] J.E. Zafrilla, M.A. Cadarso, F. Monsalve, C. de la Rúa, How carbon-friendly is nuclear energy? A Hybrid MRIO-LCA Model a Span. Facil. Environ. Sci. Technol. 48 (24) (2014) 14103–14111, <http://dx.doi.org/10.1021/es503352s>.
- [24] B. Kulisić, E. Loizou, S. Rozakis, V. Segon, Impacts of biodiesel production on Croatian economy, Energy Policy 35 (12) (2007) 6036–6045.
- [25] F. Neuwahl, A. Löschel, I. Mongelli, L. Delgado, Employment impacts of EU biofuels policy: combining bottom-up technology information and sectorial market simulations in an input-output framework, Ecol. Econ. trans-disciplinary J. Int. Soc. Ecol. Econ. (2008).
- [26] K. Mukhopadhyaya, P.J. Thomassin, Macroeconomic effects of the ethanol biofuel sector in Canada, Biomass Bioenergy 35 (7) (2011) 2822–2838, <http://dx.doi.org/10.1016/j.biombioe.2011.03.021>.
- [27] T. Silalertruksa, S.H. Gheewala, K. Hünecke, U.R. Fritsche, Biofuels and employment effects: implications for socioeconomic development in Thailand, Biomass Bioenergy 46 (2012) 409–418, <http://dx.doi.org/10.1016/j.biombioe.2012.07.019>.
- [28] E. Smeets, C. Vinyes, A. Tabeau, H. Van Meijl, C. Brink, A.G. Prins, Evaluating the Macroeconomic Impacts of Bio-based Applications in the EU, 2014. JRC Sciences and Policy Reports.
- [29] J. Song, W. Yang, Y. Higano, X.E. Wang, Introducing renewable energy and industrial restructuring to reduce GHG emission: application of a dynamic simulation model, Energy Convers. Manag. 96 (2015) 625–636.
- [30] J. Song, W. Yang, Y. Higano, X.E. Wang, Modeling the development and utilization of bioenergy and exploring the environmental economic benefits, Energy Convers. Manag. 103 (2015) 836–846.
- [31] B. Caslin, J. Finnan, L. Easson, *Miscanthus Best Practice Guidelines*, Agriculture and Food Development Authority, Agri-Food and Biosciences Institute, 2010.
- [32] D. Rizzo, L. Martin, J. Wohlfahrt, *Miscanthus spatial location as seen by farmers: a machine learning approach to model real criteria*, Biomass Bioenergy 66 (2014) 348–363, <http://dx.doi.org/10.1016/j.biombioe.2014.02.035>.
- [33] M. Kaut, R. Egging, T. Flatberg, K.T. Uggen, BLOMST – an Optimization Model

- for the Bioenergy Supply Chain. Forthcoming in Handbook of Bioenergy, Springer, 2015.
- [34] M.C. Bouwmeester, J. Oosterhaven, Specification and aggregation errors in environmentally extended input–output models, *Environ. Resour. Econ.* 56 (3) (2013) 307–335.
- [35] P. Piñero, M. Heikkinen, I. Mäenpää, E. Pongrácz, Sector aggregation bias in environmentally extended input output modeling of raw material flows in Finland, *Ecol. Econ.* 119 (2015) 217.
- [36] O. Miserque, S. Tissot, Bruart, JIndicateur des performances et des couts d'utilisation des machines agricoles: leidraad inzake prestaties en gebruikskosten van de landbouw machines. Centre de Recherches Agronomiques de Gembloux, Département Génie Rural. (1999). <https://books.google.es/books?id=r6BSnQEACAAJ>.
- [37] Eurostat -European Commission <http://ec.europa.eu/eurostat>.
- [38] IPCC, IPCC Guidelines for national greenhouse gas inventories, in: H.S. Eggleston, L. Buendia, K. Miwa, T. Ngara, K. Tanabe (Eds.), Prepared by the National Greenhouse Gas Inventories Programme, IGES, Japan, 2006, p. 2006.
- [39] G. Myhre, D. Shindell, F.-M. Bréon, W. Collins, J. Fuglestedt, J. Huang, D. Koch, J.-F. Lamarque, D. Lee, B. Mendoza, T. Nakajima, A. Robock, G. Stephens, T. Takemura, H. Zhang, Anthropogenic and natural radiative forcing, in: T.F. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S.K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex, P.M. Midgley (Eds.), *Climate Change 2013: the Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2013.
- [40] R. Parajuli, T. Dalgaard, T.L.T. Nguyen, K. Sperling, Life Cycle Assessment of Miscanthus as a fuel alternative in district heat production, in: 21st European Biomass Conference and Exhibition, Copenhagen, Denmark, 2013, pp. 388–400.